Name _____ Surname _____ Date _____

Mock test: Measurement and laboratory skills

Practice test, not graded. Duration: 45 minutes. Total: 60 marks. A calculator is allowed; there is no data sheet. Every question has four answers, A to D, and only one is correct. Write the letter of your answer in the grid at the end. A wrong answer scores the same as no answer.

1. Which of these is a derived quantity? **[2 marks]**

A) mass **B)** time **C)** volume **D)** length

2. Which unit can be used for density? **[2 marks]**

A) $\mathrm{g/cm^3}$ **B)** $\mathrm{g{\cdot}cm^3}$ **C)** $\mathrm{cm^3/g}$ **D)** $\mathrm{g/cm}$

3. One millimetre (1 mm) is equal to: **[2 marks]**

A) 1000 m **B)** 0.1 m **C)** 0.01 m **D)** 0.001 m

4. A mass of 0.36 kg is equal to: **[2 marks]**

A) 0.00036 g **B)** 360 g **C)** 3.6 g **D)** 36 000 g

5. A car travels at 72 km/h. In m/s this is: **[2 marks]**

A) 0.020 m/s **B)** 20 m/s **C)** 72 000 m/s **D)** 259 m/s

6. A density of 2.7 $\mathrm{g/cm^3}$ is equal to: **[2 marks]**

A) $0.0027\ \mathrm{kg/m^3}$
B) $27\ \mathrm{kg/m^3}$
C) $2\,700\ \mathrm{kg/m^3}$
D) $2\,700\,000\ \mathrm{kg/m^3}$

7. Which number is written in standard form? **[2 marks]**

A) 4.2×10^4 **B)** 42×10^3 **C)** 0.8×10^5 **D)** 420×10^2

8. The time 0.00052 s written in standard form is: **[2 marks]**

A) 5.2×10^4 s **B)** 5.2×10^{-3} s **C)** 52×10^{-5} s **D)** 5.2×10^{-4} s

9. A length of 2.8×10^5 mm is equal to: **[2 marks]**

A) 2.8×10^8 m **B)** 2.8×10^2 m **C)** 2.8×10^{-2} m **D)** 2.8×10^3 m

10. How many significant figures are there in 0.002050 s? **[2 marks]**

A) 2 **B)** 3 **C)** 4 **D)** 6

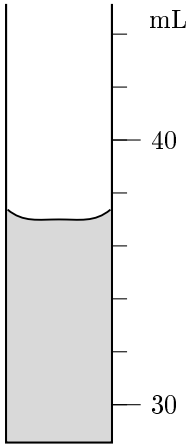
11. Which measurement has exactly 3 significant figures? **[2 marks]**

A) 0.0072 m **B)** 0.0150 kg **C)** 3.2 s **D)** 4.0×10^3 m

12. The number 0.0049851 s, rounded to 2 significant figures, is: **[2 marks]**

A) 0.0050 s **B)** 0.0049 s **C)** 0.005 s **D)** 0.00499 s

13. The diagram shows the water in a measuring cylinder. Read the volume of water. **[2 marks]**



A) 36 mL **B)** 36.5 mL **C)** 38 mL **D)** 37 mL

14. What is the uncertainty of one reading on this cylinder? **[2 marks]**

A) ± 1 mL **B)** ± 2 mL **C)** ± 0.5 mL **D)** ± 0.1 mL

15. With nothing on it, a digital balance reads 0.4 g. A stone placed on the balance makes it read 18.7 g. The correct mass of the stone is: **[2 marks]**

A) 19.1 g **B)** 18.7 g **C)** 18.3 g **D)** 0.4 g

16. A student always starts the stopwatch 0.3 s late and stops it exactly on time. This is: **[2 marks]**

A) a random error, readings too long
B) a systematic error, readings too short
C) a systematic error, readings too long
D) a random error, readings too short

17. The true mass of an object is 50.0 g. A student weighs it four times and reads 48.1 g, 48.2 g, 48.1 g and 48.2 g. The readings are: **[2 marks]**

A) accurate and precise
B) accurate but not precise
C) neither accurate nor precise
D) precise but not accurate

18. In the pendulum lab you change the length of the string and measure the time for 20 swings. Which quantity must be kept the same (a controlled variable)? **[2 marks]**

A) the length of the string
B) the time for 20 swings
C) the mass of the bob
D) the period T

19. A student times a ball rolling down a ramp five times: 2.12 s, 2.08 s, 2.71 s, 2.10 s and 2.09 s. The 2.71 s result is anomalous. The mean of the other four readings, to 3 significant figures, is: **[4 marks]**

A) 2.10 s **B)** 2.22 s **C)** 2.09 s **D)** 8.39 s

20. In the pendulum lab, a group times 20 swings and gets 36.4 s. The uncertainty in starting and stopping the stopwatch is ± 0.2 s. The time for one swing, with its uncertainty, is: **[4 marks]**

A) $T = 1.82\ \text{s} \pm 0.2\ \text{s}$
B) $T = 1.82\ \text{s} \pm 0.01\ \text{s}$
C) $T = 728\ \text{s} \pm 4\ \text{s}$
D) $T = 0.55\ \text{s} \pm 0.01\ \text{s}$

21. A pendulum has $T \approx 1.5$ s and the uncertainty in the total time is ± 0.2 s. What is the smallest number of swings you must time so that this uncertainty is less than 1% of the total time? **[4 marks]**

A) 10 swings **B)** 13 swings **C)** 14 swings **D)** 20 swings

22. In the density lab a student reads 40 mL before and 53 mL after the object is put in. Each reading has an uncertainty of half a division (0.5 mL). The relative uncertainty of the volume of the object is: **[4 marks]**

A) 3.8% **B)** 0.077% **C)** 1300% **D)** 7.7%

23. The pebble has a mass of 35.1 g and a volume of 13 cm^3. Its density is: **[4 marks]**

A) 0.37 g/cm^3 **B)** 456 g/cm^3 **C)** 2.7 kg/m^3 **D)** 2.7 g/cm^3

24. For the pebble, the relative uncertainty of the mass is 0.3% and that of the volume is 7.7%. The relative uncertainty of the density $\rho = m/V$ is: **[4 marks]**

A) 8.0% **B)** 7.4% **C)** 7.7% **D)** 0.023%

Answer grid

1.	**2.**	**3.**	**4.**	**5.**	**6.**
7.	**8.**	**9.**	**10.**	**11.**	**12.**
13.	**14.**	**15.**	**16.**	**17.**	**18.**
19.	**20.**	**21.**	**22.**	**23.**	**24.**